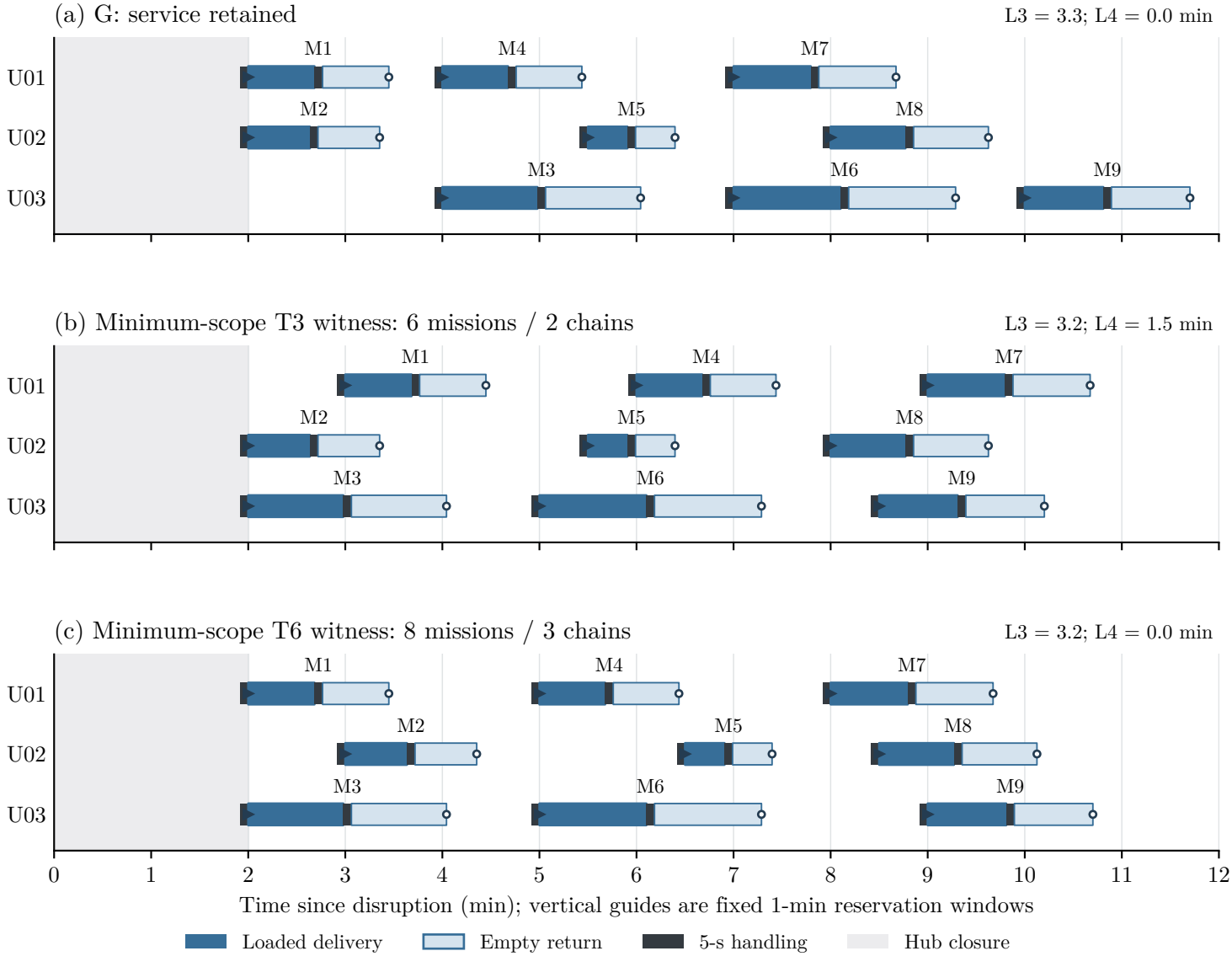
(a) G: service retained
L3 = 3.3; L4 = 0.0 min
U01
U02
U03
M1
M2
M3
M4
M5
M6
M7
M8
M9
(b) Minimum-scope T3 witness: 6 missions / 2 chains
L3 = 3.2; L4 = 1.5 min
U01
U02
U03
M1
M2
M3
M4
M5
M6
M7
M8
M9
(c) Minimum-scope T6 witness: 8 missions / 3 chains
L3 = 3.2; L4 = 0.0 min
U01
U02
U03
M1
M2
M3
M4
M5
M6
M7
M8
M9
0
1
2
3
4
5
6
7
8
9
10
11
12
Time since disruption (min); vertical guides are fixed 1-min reservation windows
Loaded delivery
Empty return
5-s handling
Hub closure